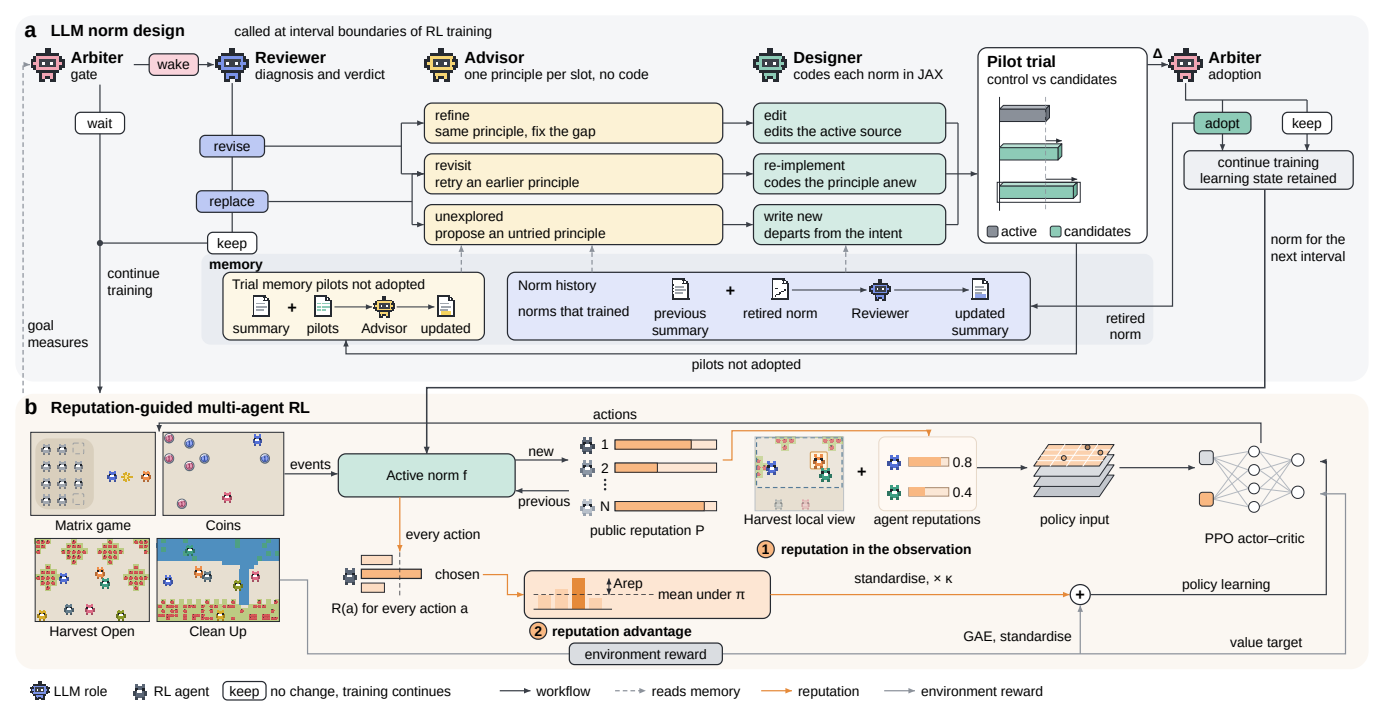

a LLM norm design
called at interval boundaries of RL training
Arbiter
gate
wake
Reviewer
diagnosis and verdict
Advisor
one principle per slot, no code
Designer
codes each norm in JAX
Pilot trial
control vs candidates
Arbiter
adoption
wait
revise
replace
keep
refine
same principle, fix the gap
revisit
retry an earlier principle
unexplored
propose an untried principle
edit
edits the active source
re-implement
codes the principle anew
write new
departs from the intent
active
candidates
adopt
keep
continue training
learning state retained
norm for the next interval
memory
Trial memory pilots not adopted
summary
pilots
Advisor
updated
Norm history
norms that trained
previous summary
retired norm
Reviewer
updated summary
retired norm
continue training
goal measures
pilots not adopted
b Reputation-guided multi-agent RL
actions
Matrix game
Coins
events
Active norm f
new
previous
1
2
N
public reputation P
Harvest local view
0.8
0.4
agent reputations
policy input
PPO actor–critic
1 reputation in the observation
every action
chosen
R(a) for every action a
Harvest Open
Clean Up
Arep
mean under π
2 reputation advantage
standardise, × κ
policy learning
GAE, standardise
value target
environment reward
LLM role
RL agent
keep no change, training continues
workflow
reads memory
reputation
environment reward